APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Algebra & Quadratic Equations - Unit 3 Assignment

SUBJECT: Mathematics (MATH101)

CLASS & SECTION: Class 10 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 95 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Solve all exercises from 3.1 to 3.5 showing all working steps clearly with quadratic formulas and graphs.

1. Executive Overview & Problem Context

Unit 3 investigates quadratic polynomial equations, algebraic factoring techniques, completing the square, quadratic formula derivations, and geometric interpretations of parabolic functions. This dossier provides the comprehensive step-by-step solution manual for exercises 3.1 through 3.5.

2. Core Analysis, Solutions & Systematic Working

- **Exercise 3.1: Identification & Standard Form Conversion:**

  Standard form: $ax^2 + bx + c = 0$ (a "` 0). Transformed $(2x - 1)(x - 3) = (x + 5)(x - 1)$ -> $2x^2 - 7x + 3 = x^2 + 4x - 5$ -> $x^2 - 11x + 8 = 0$. Here a = 1, b = -11, c = 8.

- **Exercise 3.2: Factorization & Middle Term Splitting:**

  Solve: " $2x^2 + 7x + 5$" 2 = 0. Product ac = " 2 × 5" 2 = 10; sum b = 7. Splitting factors: 5 and 2. " $2x^2 + 2x + 5x + 5$" 2 = 0 -> " 2 x(x + " 2) + 5(x + " 2) = 0 -> (x + " 2)(" 2 x + 5) = 0. Roots: x = -" 2 and x = -5/" 2.

- **Exercise 3.3: Method of Completing the Square:**

  Solve: $2x^2 - 5x + 3 = 0$. Divide by 2: $x^2 - (5/2)x = -3/2$. Add $[5/4]^2$ to both sides: $[x - 5/4]^2 = 25/16 - 24/16 = 1/16$. Take square root: x - 5/4 = ± 1/4. Roots: x = 6/4 = 3/2 and x = 4/4 = 1.

- **Exercise 3.4 & 3.5: Real-World Word Problems & Nature of Roots:**

  Problem: The diagonal of a rectangular field is 60m more than the shorter side. If the longer side is 30m more than the shorter side, find field sides. Let shorter side = x. Longer side = x + 30. Diagonal = x + 60. By Pythagoras: $(x + 60)^2 = x^2 + (x + 30)^2$. Expands to: $x^2 - 60x - 2700 = 0$ -> $(x - 90)(x + 30) = 0$. Since length > 0, x = 90m, longer side = 120m, diagonal = 150m.

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3. Detailed Data Matrix & Systematic Findings

Exercise Problem	Algebraic Equation	Solution Method	Computed Roots	Verification
Ex 3.1 #4	x^2 - 11x + 8 = 0	Quadratic Formula	x = (11 ± " 89) / 2	Discriminant D = 89 > 0
Ex 3.2 #2	" 2 x^2 + 7x + 5" 2 = 0	Middle Term Splitting	x = -" 2, -5/" 2	Product of roots = 5
Ex 3.3 #1	2x^2 - 5x + 3 = 0	Completing the Square	x = 1.5, 1.0	Exact integer fractional roots
Ex 3.4 #3	2x^2 + kx + 3 = 0	Equal Roots Condition (D=0)	k = ± 2" 6	k^2 - 24 = 0 -> k = ±" 24
Ex 3.5 #2	x^2 - 60x - 2700 = 0	Factoring Dimensions	x = 90 meters	Sides: 90m by 120m, Diag 150m

4. Student Conclusions & Reflective Synthesis

Mastery of quadratic equations synthesizes geometric coordinate graphing, algebraic factorization, and practical optimization modeling required across STEM disciplines.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Problem Solving & Structured Working: 18 / 20

Analytical Depth & Research Methodology: 28 / 30

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding mathematical rigor! Clear working steps, factorization proofs, and parabola vertex calculations."

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